Entropy–Complexity Based Clustering of ARMA Simulated Time Series Models

M. H. M. R. S. Dilhani *[1], Alejandro C. Frery[1], and A. A. Rey[2]

[1] *School of Mathematics and Statistics, Victoria University of Wellington, New Zealand*
[2] *Laboratorio de Investigación y Desarrollo Experimental en Computación (LIDEC),, Instituto de Tecnología e Ingeniería, Universidad Nacional de Hurlingham (UNAHUR), Argentina*

Abstract

Ordinal pattern analysis has emerged as a promising alternative to analysing time series data, providing a robust and computationally efficient approach. It involves converting time series data into a sequence of symbols that represent ordering relationships among data points within specific time windows. Then, the Shannon entropy and the Statistical Complexity are computed from the histogram of symbols. The method detects small differences between models and sample sizes. We used ordinal pattern analysis to analyse simulated time series under Autoregressive (AR), Moving Average (MA), and Autoregressive Moving Average (ARMA) models.

This study presents a simulation-based experiment into time series using ordinal pattern analysis. The primary research focus was on identifying the distinguishing characteristics of AR, MA, and ARMA models through ordinal patterns features. Simulated time series were generated with series length $n = 1000$, using parameter values chosen to satisfy the stationarity and invertibility conditions of the model. The experimental design included several variants for each AR, MA, and ARMA model type, with each model representing a distinct parameter setting (Table 1). For ARMA models, the parameters α_1, α_2 correspond to the autoregressive model meanwhile β_1 and β_2 correspond to the moving average model. These parameters define the model structure and directly affect the time series dynamics examined in this study. To ensure model accuracy, 100 independent replications were performed for each model configuration.

Ordinal patterns were extracted to compute entropy and complexity. Features were analysed in the entropy–complexity plane for model discrimination. ARMA$(1, 1)$ shows two distinct clusters. AR coefficients -0.8 and 0.8 produce lower entropy and higher complexity. Coefficients 0.1 and -0.1 yield higher entropy and lower complexity. ARMA$(2, 2)$ shows overlapping groups across all four models. Clear separation appears by sample size, with larger samples forming more stable clusters, as seen in Figure 1.

The findings highlight the effectiveness of feature-based ordinal pattern analysis of time series, which uses Shannon entropy and complexity to characterize intrinsic differences among models. The simulation framework further validates the method's reliability under varying structural and stochastic configurations. This research establishes a foundation for applying ordinal pattern analysis to unsupervised time series grouping, with potential applications in data-rich domains such as finance, engineering, and bioinformatics.

*Contact: rasika.dilhani@vuw.ac.nz

Type	**Model**	α_1	α_2	β_1	β_2
AR(1)	$\mathrm{AR}(1)_{\mathrm{M}_1}$	0.8			
	$\mathrm{AR}(1)_{\mathrm{M}_2}$	0.1			
	$\mathrm{AR}(1)_{\mathrm{M}_3}$	−0.8			
	$\mathrm{AR}(1)_{\mathrm{M}_4}$	−0.1			
AR(2)	$\mathrm{AR}(2)_{\mathrm{M}_1}$	0.1	0.8		
	$\mathrm{AR}(2)_{\mathrm{M}_2}$	−0.8	0.1		
	$\mathrm{AR}(2)_{\mathrm{M}_3}$	0.1	−0.8		
	$\mathrm{AR}(2)_{\mathrm{M}_4}$	−0.8	−0.1		
MA(1)	$\mathrm{MA}(1)_{\mathrm{M}_1}$			0.8	
	$\mathrm{MA}(1)_{\mathrm{M}_2}$			0.1	
	$\mathrm{MA}(1)_{\mathrm{M}_3}$			−0.8	
	$\mathrm{MA}(1)_{\mathrm{M}_4}$			−0.1	
MA(2)	$\mathrm{MA}(2)_{\mathrm{M}_1}$			0.1	0.8
	$\mathrm{MA}(2)_{\mathrm{M}_2}$			−0.8	0.1
	$\mathrm{MA}(2)_{\mathrm{M}_3}$			0.1	−0.8
	$\mathrm{MA}(2)_{\mathrm{M}_4}$			−0.8	−0.1
ARMA(1, 1)	$\mathrm{ARMA}(1,1)_{\mathrm{M}_1}$	0.8		0.8	
	$\mathrm{ARMA}(1,1)_{\mathrm{M}_2}$	0.1		0.1	
	$\mathrm{ARMA}(1,1)_{\mathrm{M}_3}$	−0.8		−0.8	
	$\mathrm{ARMA}(1,1)_{\mathrm{M}_4}$	−0.1		−0.1	
ARMA(2, 2)	$\mathrm{ARMA}(2,2)_{\mathrm{M}_1}$	0.1	0.8	0.1	0.8
	$\mathrm{ARMA}(2,2)_{\mathrm{M}_2}$	−0.8	0.1	−0.8	0.1
	$\mathrm{ARMA}(2,2)_{\mathrm{M}_3}$	0.1	−0.8	0.1	−0.8
	$\mathrm{ARMA}(2,2)_{\mathrm{M}_4}$	−0.8	−0.1	−0.8	−0.1

Table 1: Models and parameter sets for AR(1), AR(2), MA(1), MA(2), ARMA(1, 1), and ARMA(2, 2). Empty cells indicate omitted parameters.

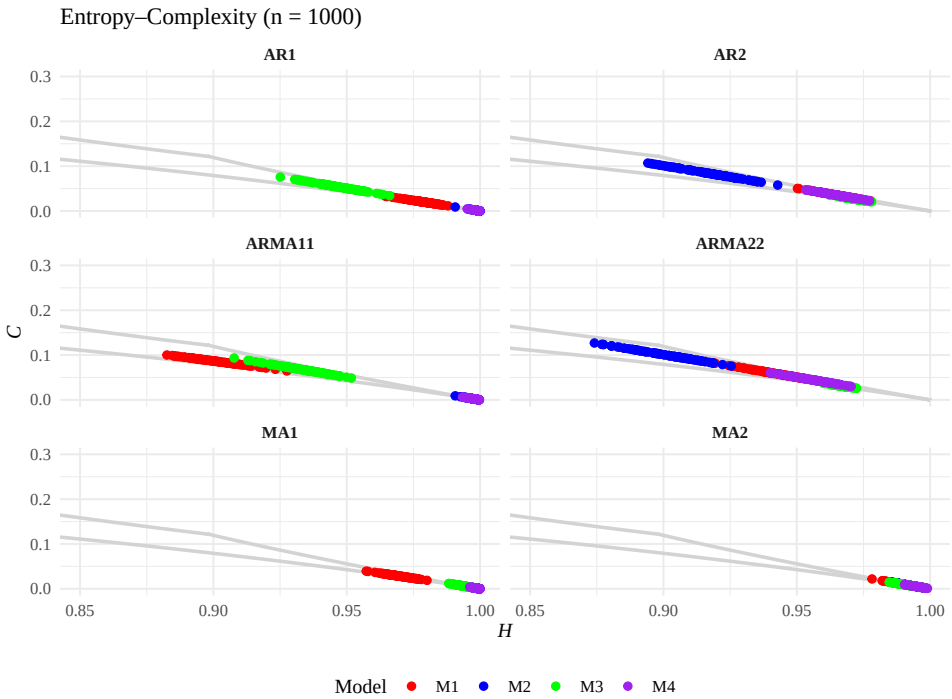

Figure 1: $H \times C$ plane for new model n1000